Chapter 5
Lists (cont) - Linked List

Data Structures and Algorithms

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Outcomes

- **L.O.2.1** - Depict the following concepts: (a) array list and linked list, including single link and double links, and multiple links.
- **L.O.2.2** - Describe storage structures by using pseudocode for: (a) array list and linked list, including single link and double links, and multiple links.
- **L.O.2.3** - List necessary methods supplied for list and describe them using pseudocode.
- **L.O.2.4** - Implement list using C/C++.

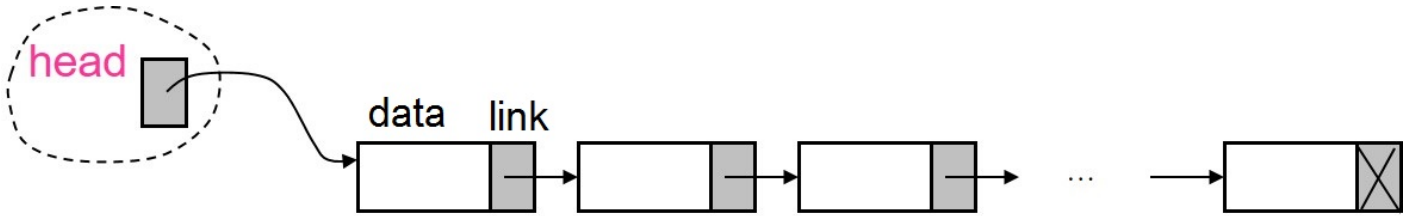
Outcomes

- **L.O.2.5** - Use list for problems in real-life, and choose an appropriate implementation type (array vs. link).
- **L.O.2.6** - Analyze the complexity and develop experiment (program) to evaluate the efficiency of methods supplied for list.
- **L.O.8.4** - Develop recursive implementations for methods supplied for the following structures: list.
- **L.O.1.2** - Analyze algorithms and use Big-O notation to characterize the computational complexity of algorithms composed by using the following control structures: sequence, branching, and iteration (not recursion).

Contents

Singly linked list

Linked List



Hình: Singly Linked List

```
list // Linked Implementation of List
  head <pointer>
  count <integer> // number of elements (optional)
end list
```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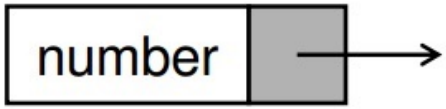
Nodes

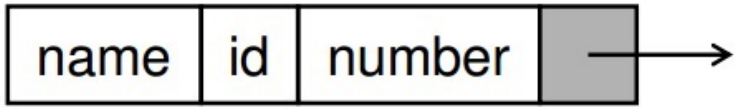
The elements in a linked list are called nodes.
A node in a linked list is a structure that has at least two fields:

- the data,
- the address of the next node.

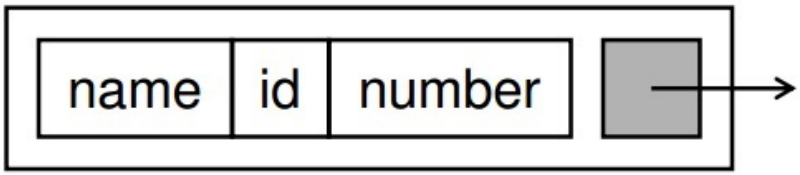
A node with one data field



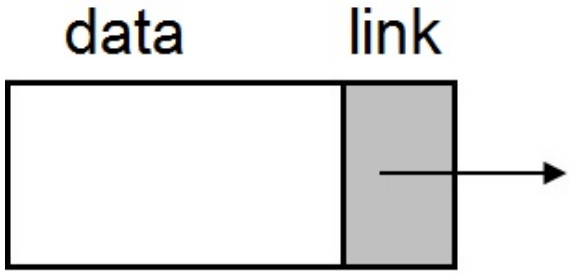
A node with three data fields



A node with one structured data field



Nodes

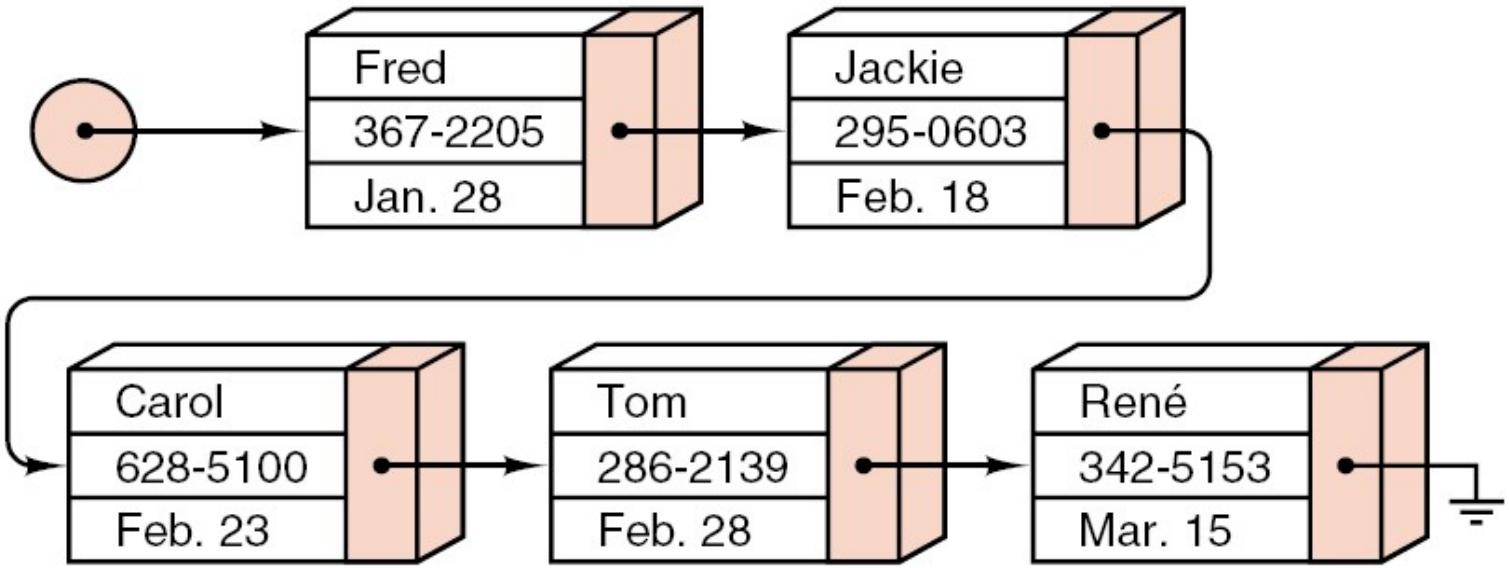


Hình: Linked list node structure

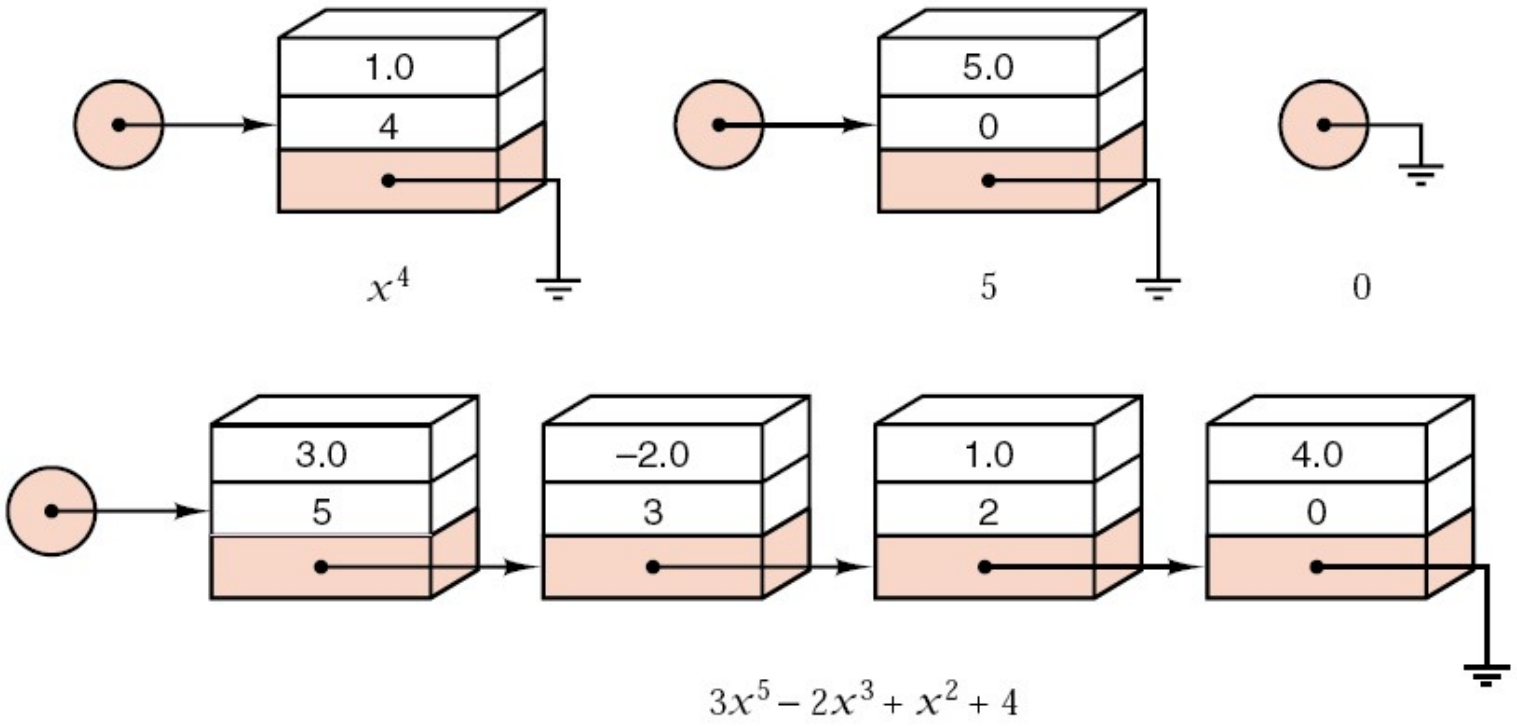
```
node
   data <dataType>
   link <pointer>
end node
```

```
// General dataType:
dataType
    key <keyType>
    field1 <...>
    field2 <...>
    ...
    fieldn <...>
end dataType
```


Example


Fred
367-2205
Jan. 28
Jackie
295-0603
Feb. 18
Carol
628-5100
Feb. 23
Tom
286-2139
Feb. 28
René
342-5153
Mar. 15

Example



Hình: List representing polynomial

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Implementation in C++

Example

```
node
  data <dataType>
  link <pointer>
end node
```

```
struct Node {
    int data;
    Node *link;
};
```


Implementation in C++

Example

```
struct Node {
    int data;
    Node *link;
};
```

```
struct Node {
    float data;
    Node *link;
};
```

```
template <class ItemType>
struct Node {
    ItemType data;
    Node<ItemType> *link;
};
```

Node implementation in C++

```
class Node {
  public:
    int data;
    Node *link;
    Node(){
      this->link = NULL;
    }

    Node(int data){
      this->data = data;
      this->link = NULL;
    }
};
```


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Linked list implementation in C++

```
class LinkedList {
  Node *head;
  int count;
public:
  LinkedList();
  ~LinkedList();
};
```

```
list
  head <pointer>
  count <integer>
end list
```


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Linked list operations

- Create an empty linked list
- Insert a node into a linked list
- Delete a node from a linked list
- Traverse a linked list
- Destroy a linked list

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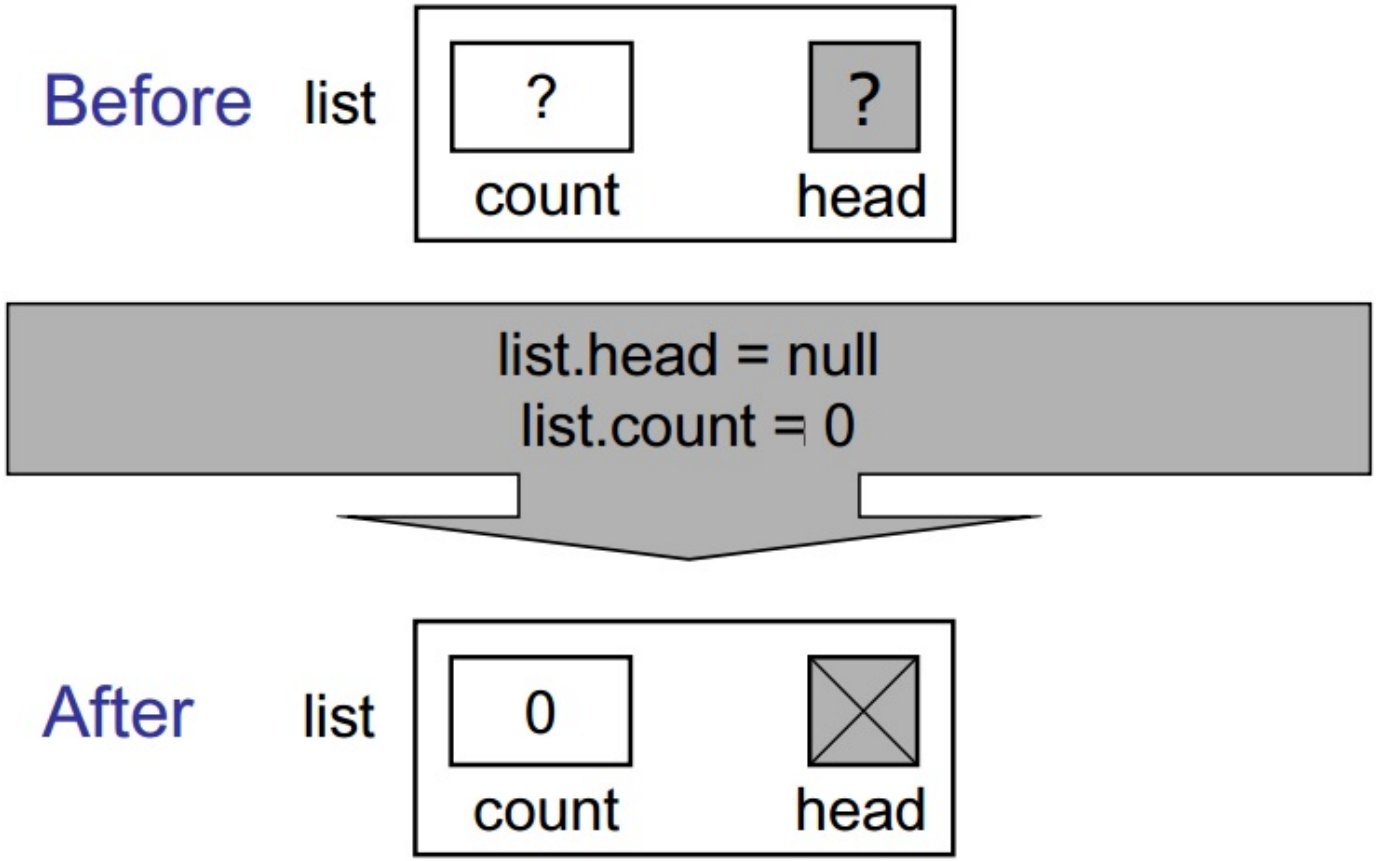
Create an empty linked list



Create an empty linked list

Algorithm createList(ref list <metadata>)
Initializes metadata for a linked list
Pre: `list` is a metadata structure passed by reference
Post: metadata initialized
list.head = null
list.count= 0
return
End createList

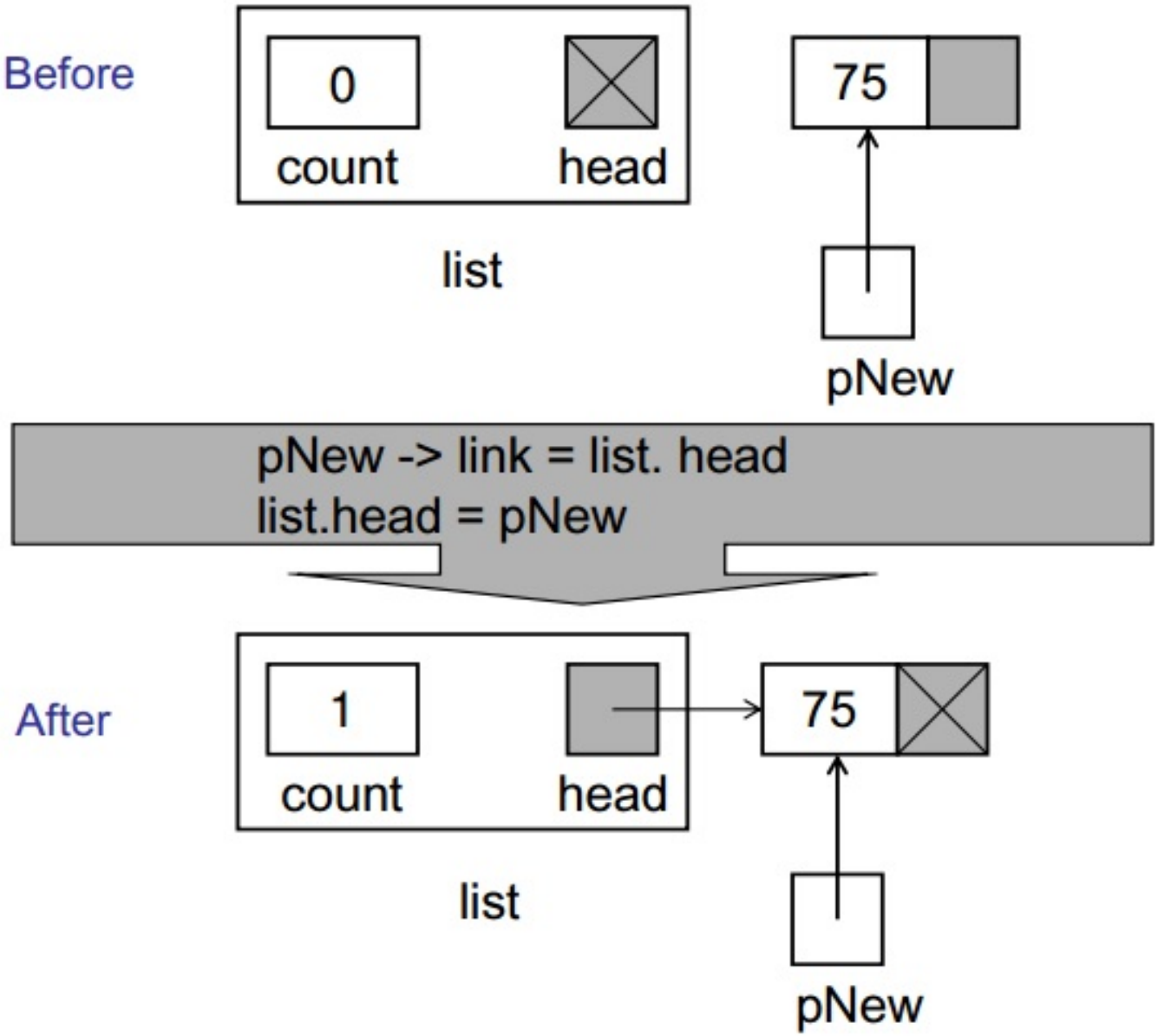
Create an empty linked list

```
class LinkedList {
  Node *head;
  int count;

  public:
    LinkedList();
    ~LinkedList();
};


LinkedList::LinkedList(){
  this->head = NULL;
  this->count = 0;
}
```


Insert a node into a linked list

1. Allocate memory for the new node and set up data.
2. Locate the pointer p in the list, which will point to the new node:
   - If the new node becomes the first element in the List: `p is list.head`.
   - Otherwise: `p is pPre->link`, where `pPre` points to the predecessor of the new node.
3. Point the new node to its successor.
4. Point the pointer p to the new node.

Insert into an empty linked list



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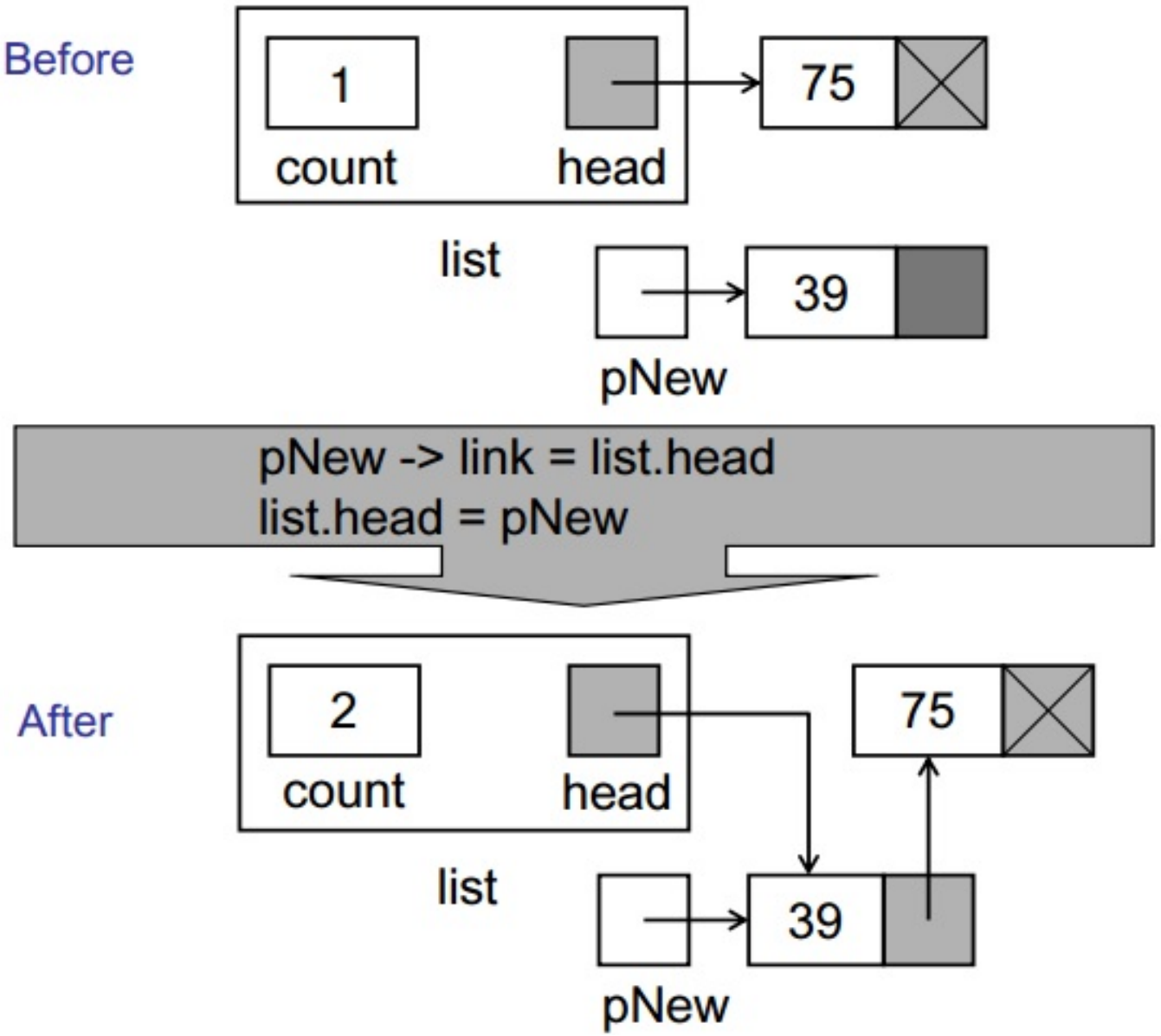
Insert at the beginning



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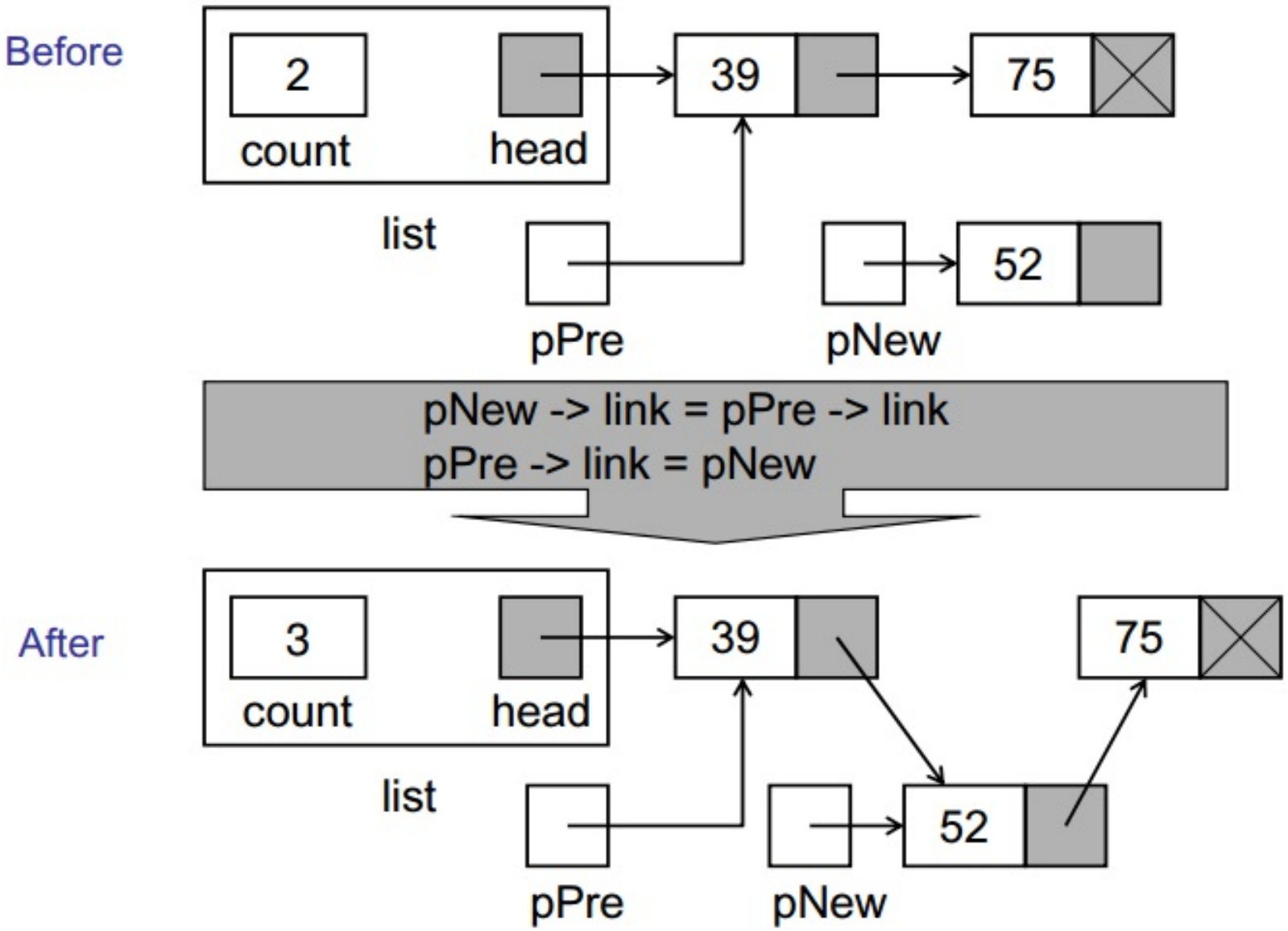
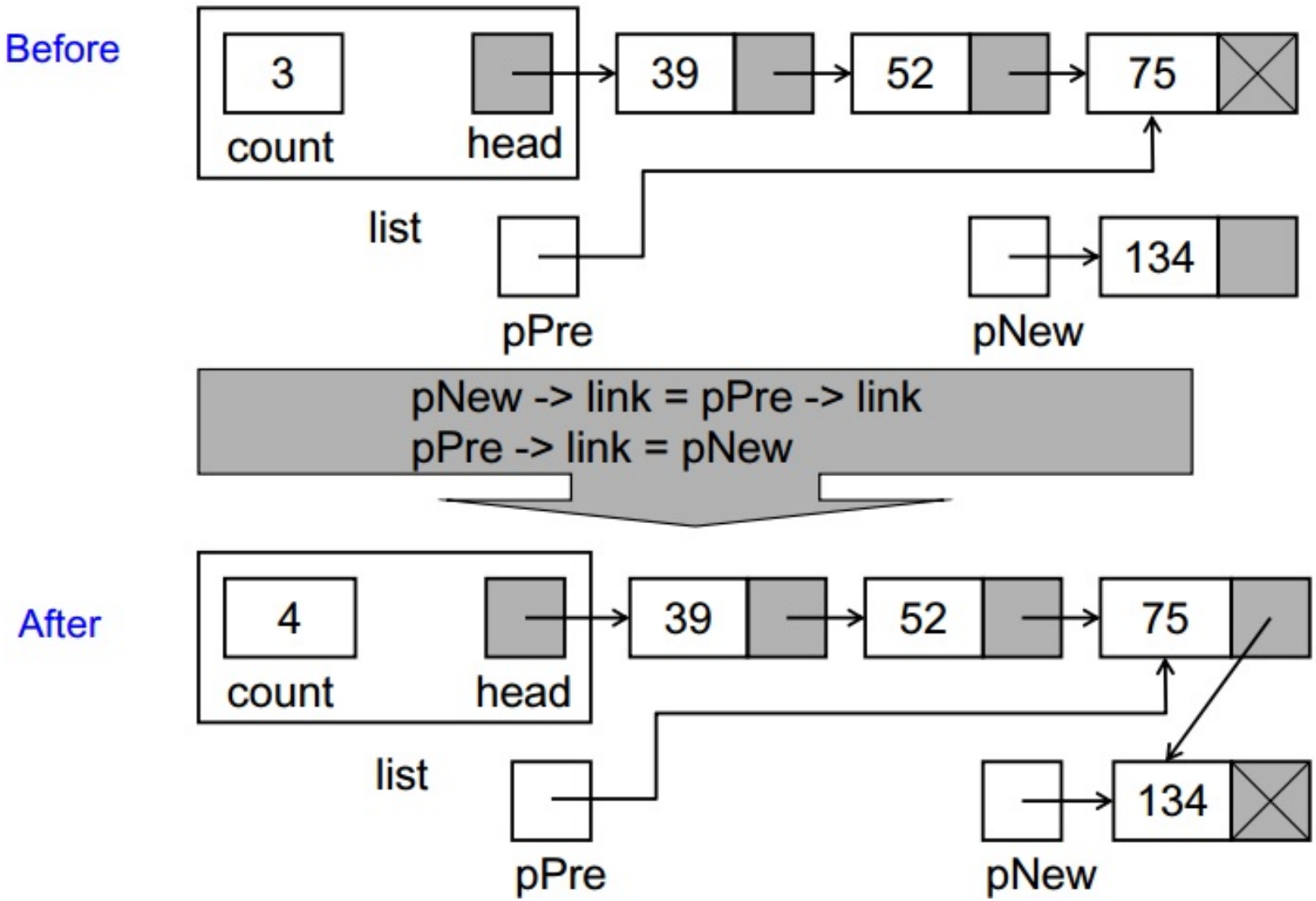
Insert in the middle



Insert at the end



Insert a node into a linked list

- Insertion is successful when allocation memory for the new node is successful.
- There is no difference between insertion at the beginning of the list and insertion into an empty list.

```
pNew->link = list.head
list.head = pNew
```

- There is no difference between insertion in the middle and insertion at the end of the list.

```
pNew->link = pPre->link
pPre->link = pNew
```


Insert a node into a linked list

Algorithm insertNode(ref list <metadata>,
val pPre <node pointer>,
val dataIn <dataType>)
Inserts data into a new node in the linked list.

Pre: `list` is metadata structure to a valid list
`pPre` is pointer to data's logical predecessor
`dataIn` contains data to be inserted
Post: data have been inserted in sequence
Return true if successful, false if memory overflow

Insert a node into a linked list

```
allocate(pNew)
if memory overflow then
    return false
end
pNew -> data = dataIn
if pPre = null then
    // Adding at the beginning or into empty list
    pNew -> link = list.head
    list.head = pNew
else
    // Adding in the middle or at the end
    pNew -> link = pPre -> link
    pPre -> link = pNew
end
list.count = list.count + 1
return true
End insertNode
```


Insert a node into a linked list

```
int LinkedList::InsertNode(Node*pPre,
        int value) {
  Node *pNew = new Node();
  if (pNew == NULL)
    return 0;
  pNew->data = value;
  if (pPre== NULL){
    pNew->link = this->head;
    this->head = pNew;
  } else {
    pNew->link = pPre->link;
    pPre->link = pNew;
  }
  this->count++;
  return 1;
}
```


Delete a node from a linked list

1. Locate the pointer p in the list which points to the node to be deleted (pLoc will hold the node to be deleted).
   - If that node is the first element in the List: `p is list.head`.
   - Otherwise: `p is pPre->link`, where `pPre` points to the predecessor of the node to be deleted.
2. `p` points to the successor of the node to be deleted.
3. Recycle the memory of the deleted node.

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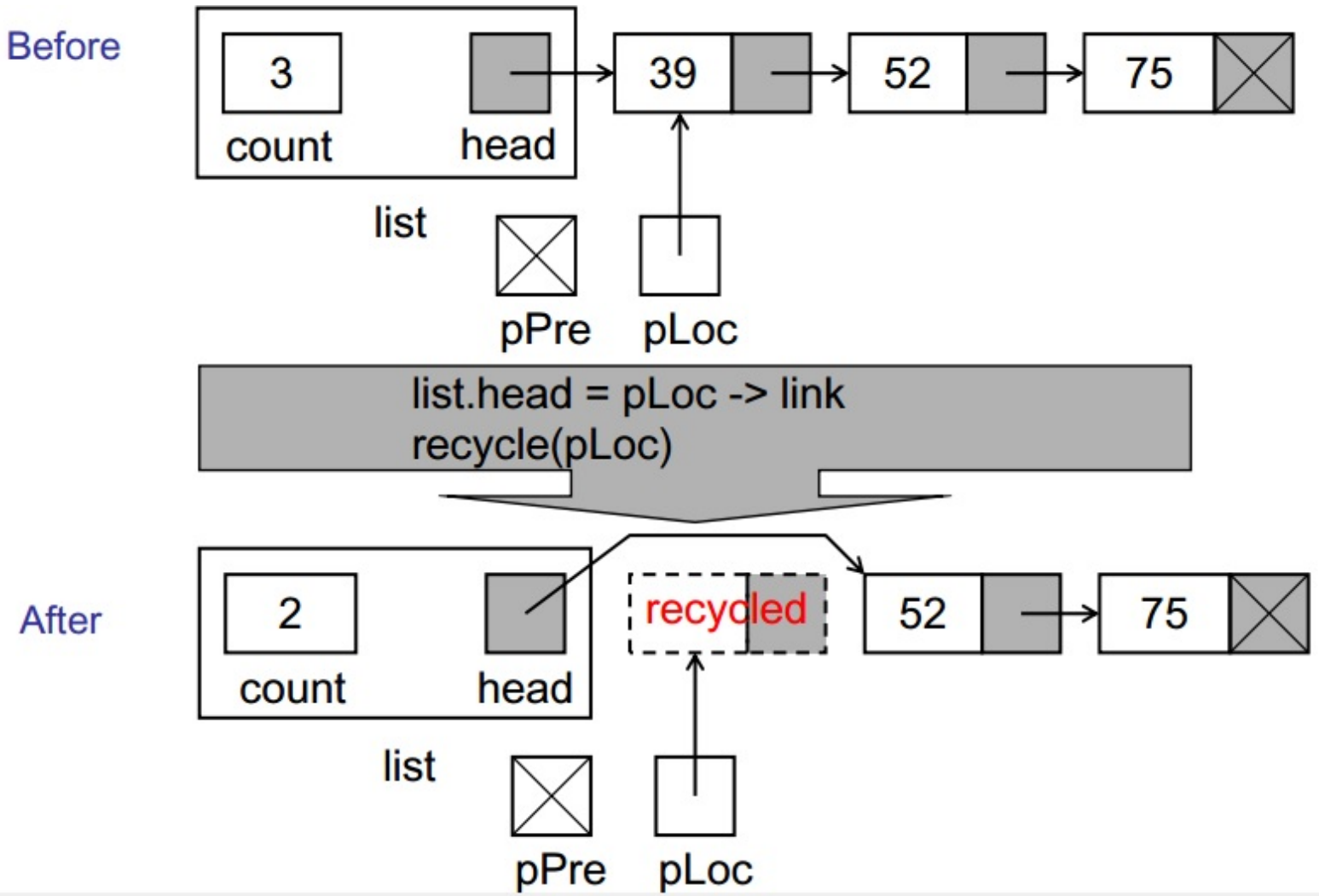
Delete first node



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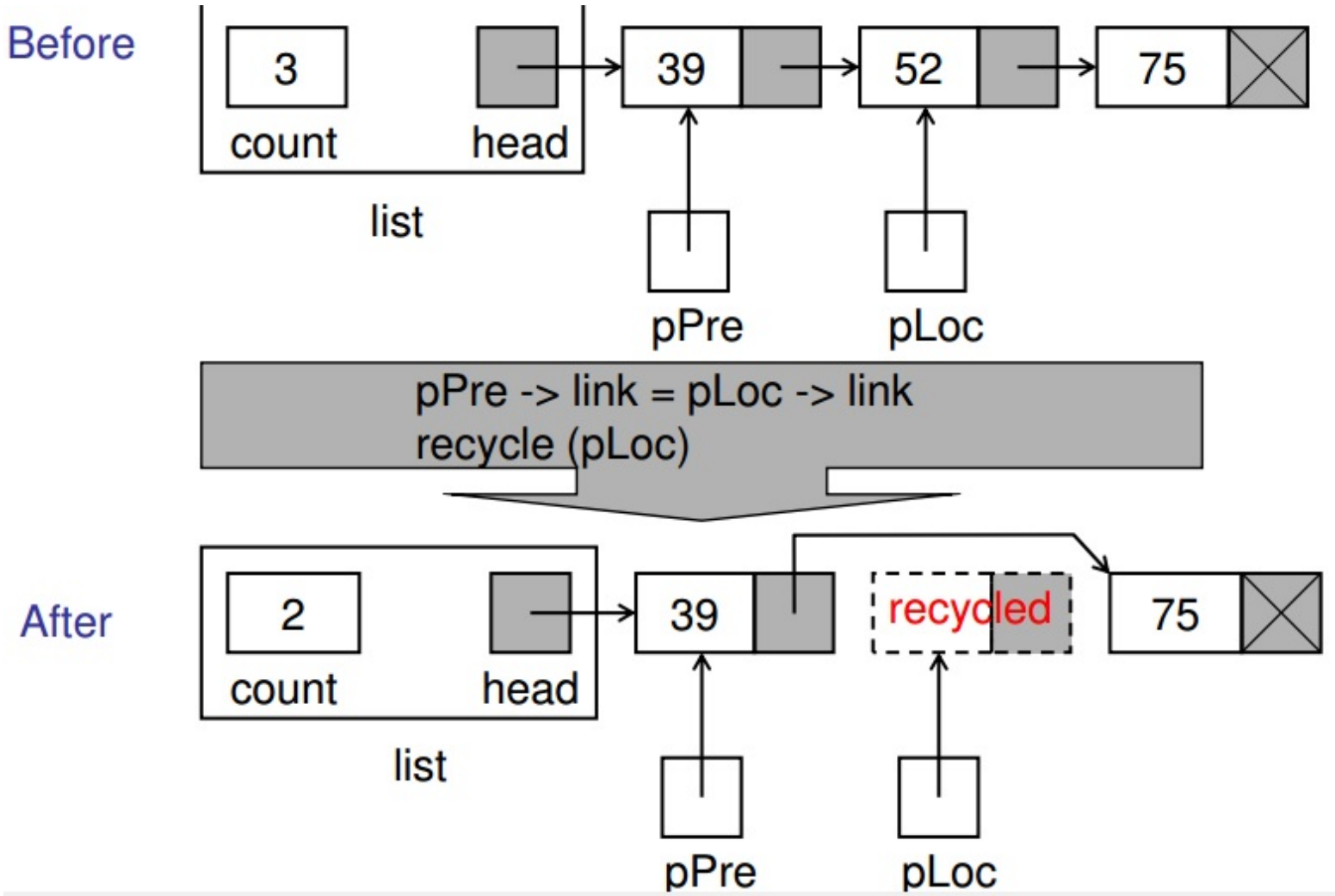
General deletion case



Delete a node from a linked list

- Removal is successful when the node to be deleted is found.
- There is no difference between deleting the node from the beginning of the list and deleting the only node in the list.

```
list.head = pLoc->link
recycle(pLoc)
```

- There is no difference between deleting a node from the middle and deleting a node from the end of the list.

```
pPre->link = pLoc->link
recycle(pLoc)
```


Delete a node from a linked list

Algorithm deleteNode(ref list <metadata>,
val pPre <node pointer>,
val pLoc <node pointer>,
ref dataOut <dataType>)
Deletes data from a linked list and returns it to calling module.

Pre: `list` is metadata structure to a valid list
`pPre` is a pointer to predecessor node
`pLoc` is a pointer to node to be deleted
`dataOut` is variable to receive deleted data
Post: data have been deleted and returned to caller

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Delete a node from a linked list

```
dataOut = pLoc -> data
if pPre = null then
    // Delete first node
    list.head = pLoc -> link
else
    // Delete other nodes
    pPre -> link = pLoc -> link
end
list.count = list.count - 1
recycle (pLoc)
return
End deleteNode
```


Delete a node from a linked list

```
int LinkedList::DeleteNode(Node *pPre, Node *pLoc) {
  int result = pLoc->data;

  if (pPre== NULL){
    this->head = pLoc->link;
  } else {
    pPre->link = pLoc->link;
  }

  this->count--;
  delete pLoc;
  return result;
}
```

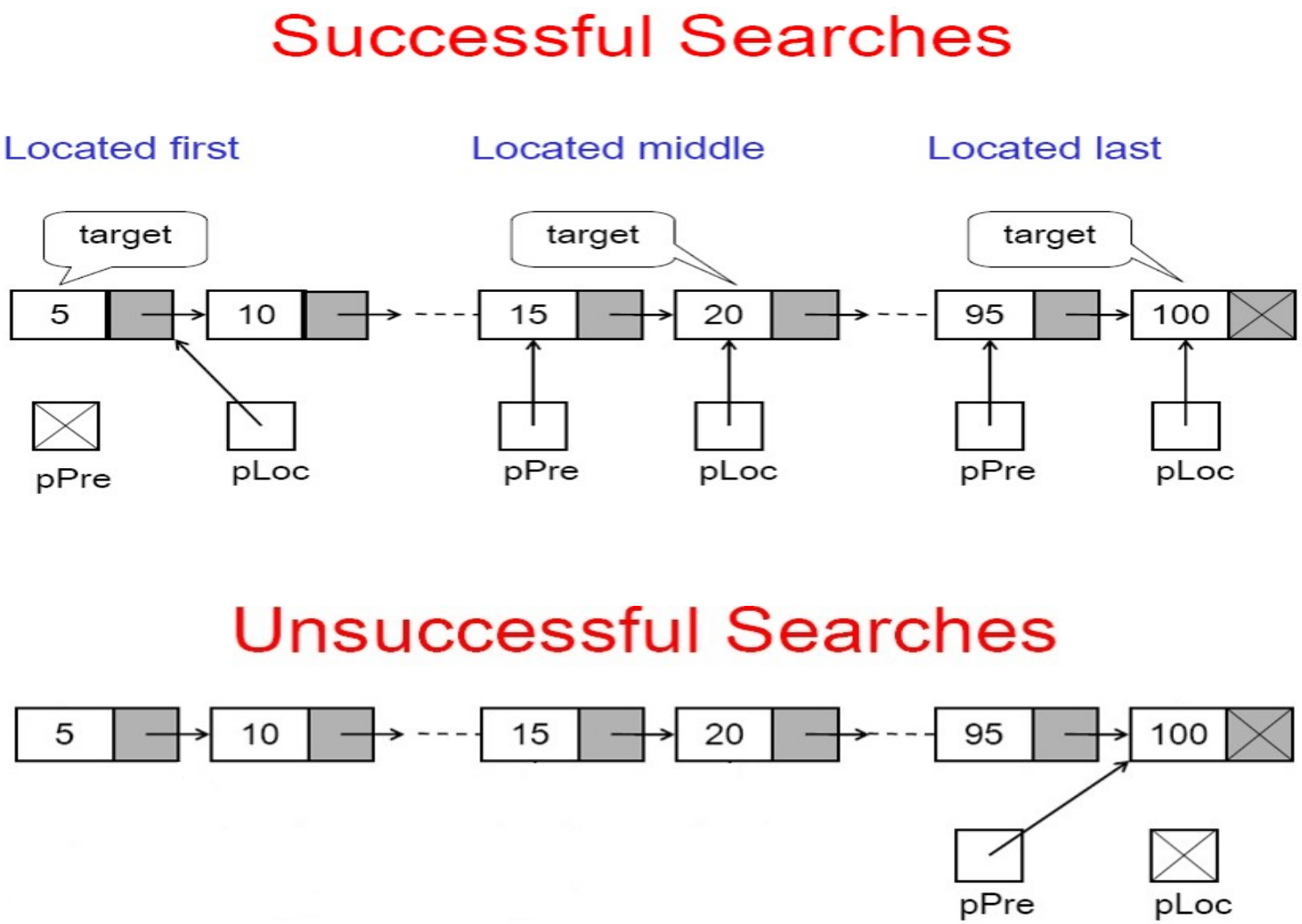

Searching in a linked list

- Sequence Search has to be used for the linked list.
- Function Search of List ADT:

```
<ErrorCode> Search (val target <dataType>,
        ref pPre <pointer>, ref pLoc <pointer>)
```

Searches a node and returns a pointer to it if found.

Searching in a linked list



Searching in a linked list

Algorithm Search(val target <dataType>,
ref pPre <node pointer>,
ref pLoc <node pointer>)
Searches a node in a singly linked list and return a pointer to it if found.

Pre: `target` is the value need to be found

Post: `pLoc` points to the first node which is equal target, or is NULL if not found.
`pPre` points to the predecessor of the first node which is equal target, or points to the last node if not found.

Return `found` or `notFound`

Searching in a linked list

```
pPre = NULL
pLoc = list.head
while (pLoc is not NULL) AND
(target != pLoc ->data) do
    pPre = pLoc
    pLoc = pLoc ->link
end
if pLoc is NULL then
    return notFound
else
    return found
end
End Search
```


Searching in a linked list

```
int LinkedList::Search(int value,
        Node* &pPre, Node* &pLoc){
  pPre = NULL;
  pLoc = this->head;
  while (pLoc != NULL && pLoc->data != value){
    pPre = pLoc;
    pLoc = pLoc->link;
  }
  return (pLoc != NULL);
  // found: 1; notfound: 0
}
```


Traverse a linked list

Traverse module controls the loop: calling a user-supplied algorithm to process data

Algorithm Traverse(ref <void> process (ref Data <DataType>))
Traverses the list, performing the given operation on each element.
Pre: `process` is user-supplied
Post: The action specified by `process` has been performed on every element in the list, beginning at the first element and doing each in turn.

```
pWalker = list.head
while pWalker not null do
    process(pWalker -> data)
    pWalker = pWalker -> link
end
End Traverse
```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Traverse a linked list

```
void LinkedList::Traverse() {
  Node *p = head;
  while (p != NULL){
    p->data++; // process data here!!!
    p = p->link;
  }
}

void LinkedList::Traverse2(int *&visit){
  Node *p = this->head;
  int i = 0;
  while (p != NULL && i < this->count){
    visit[i] = p->data;
    p = p->link;
    i++;
  }
}
```


Destroy a linked list

Algorithm destroyList (val list <metadata>)
Deletes all data in list.

Pre: `list` is metadata structure to a valid list
Post: all data deleted
while *list.head not null* **do**
    dltPtr = list.head
    list.head = this.head -> link
    recycle (dltPtr)
end
No data left in list. Reset metadata
list.count = 0
return
End destroyList

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Destroy a linked list

```
void LinkedList::Clear(){
  Node *temp;
  while (this->head != NULL){
    temp = this->head;
    this->head = this->head->link;
    delete temp;
  }
  this->count = 0;
}


LinkedList::~LinkedList(){
  this->Clear();
}
```


Linked list implementation in C++

```
class LinkedList{
  Node* head;
  int count;
protected:
  int InsertNode(Node* pPre,int value);

  int DeleteNode(Node* pPre, Node* pLoc);

  int Search(int value, Node* &pPre, Node* &pLoc);

};
```


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Linked list implementation in C++

```
class LinkedList{

protected:
  // ...

public:
  LinkedList();
  ~LinkedList();

  void InsertFirst(int value);
  void InsertLast(int value);
  int InsertItem(int value, int position);

  void DeleteFirst();
  void DeleteLast();
  int DeleteItem(int postion);
  int GetItem(int position, int &dataOut);

  void Traverse();
  LinkedList* Clone();
  void Print2Console();
  void Clear();
};
```


Linked list implementation in C++

How to use Linked List data structure?

```
int main(int argc, char* argv[]) {
  LinkedList* myList = new LinkedList();
  myList->InsertFirst(15);
  myList->InsertFirst(10);
  myList->InsertFirst(5);
  myList->InsertItem(18,3);
  myList->InsertLast(25);
  myList->InsertItem(20,3);
  myList->DeleteItem(2);
  cout << "List 1:" << endl;
  myList->Print2Console();
```


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Linked list implementation in C++

How to use Linked List data structure?

```
    // ...

    int value;
    LinkedList* myList2 = myList->Clone();
    cout << "List 2:" << endl;
    myList2->Print2Console();
    myList2->GetItem(1, value);
    cout << "Value at position 1: " << value;

    delete myList;
    delete myList2;
    return 1;
}
```


Sample Solution: Insert

```
int LinkedList<List_ItemType>::InsertItem(
        int value, int position) {

  if (position < 0 || position > this->count)
    return 0;

  Node *newPtr, *pPre;
  newPtr = new Node();
  if (newPtr == NULL)
    return 0;

  newPtr->data = value;

  if (head == NULL) {
    head = newPtr;
    newPtr->link = NULL;
  } else if (position == 0) {
    newPtr->link = head;
    head = newPtr;
  }
```

Sample Solution: Insert

```
    else {
        // Find the position of pPre
        pPre = this->head;

        for (int i = 0; i < position-1; i++)
            pPre = pPre->link;

        // Insert new node
        newPtr->link = pPre->link;
        pPre->link = newPtr;
    }

    this->count++;
    return 1;
}
```


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Sample Solution: Delete

```
int LinkedList::DeleteItem(int position){
  if (position < 0 || position > this->count)
    return 0;

  Node *dltPtr, *pPre;

  if (position == 0) {
    dltPtr = head;
    head = head->link;
  } else {
    pPre= this->head;
    for (int i = 0; i < position -1; i++)
      pPre = pPre->link;
    dltPtr = pPre->link;
    pPre->link = dltPtr->link;
  }

  delete dltPtr;
  this->count--;
  return 1;
}
```

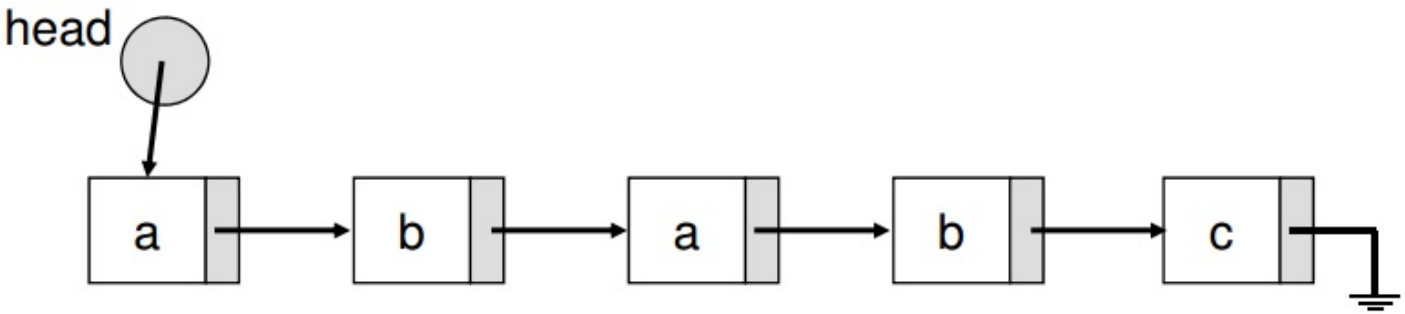
Sample Solution: Clone

```
LinkedList* LinkedList::Clone(){
  LinkedList* result = new LinkedList();

  Node* p = this->head;

  while (p != NULL) {
    result->InsertLast(p->data);
    p = p->link;
  }

  result->count = this->count;
  return result;
}
```

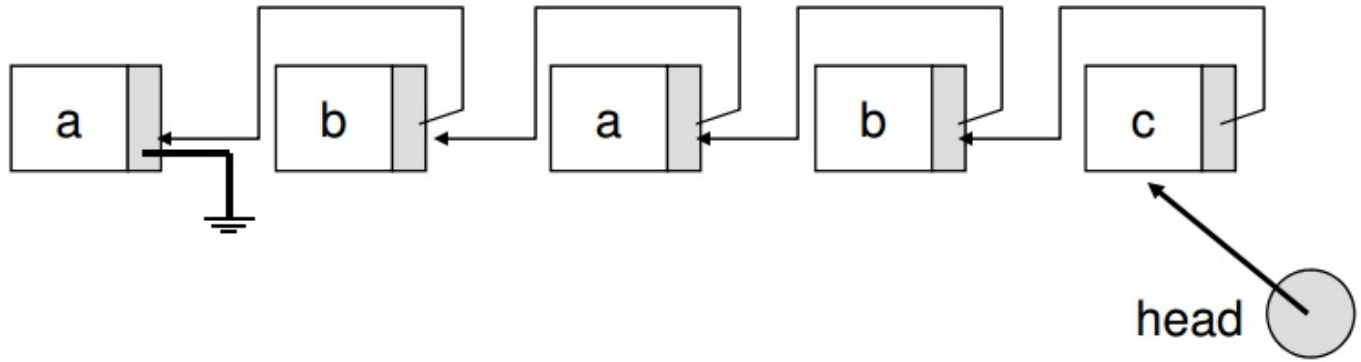

Reverse a linked list

Exercise

```
void LinkedList::Reverse(){
   // ...
}
```

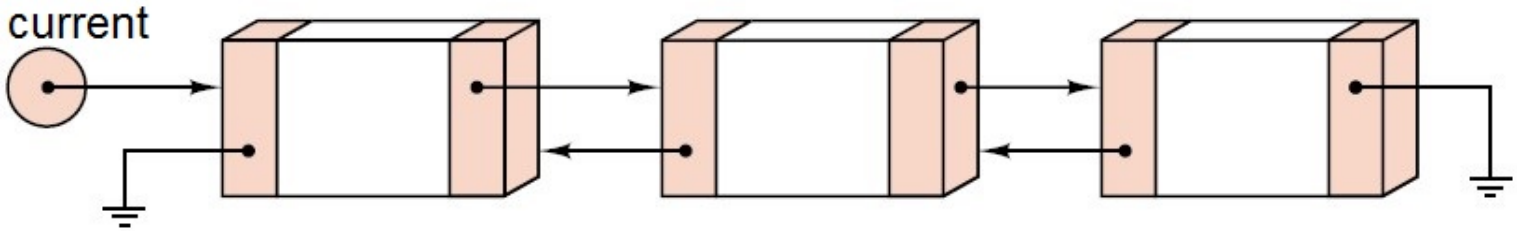


Result:



Other linked lists

Doubly Linked List

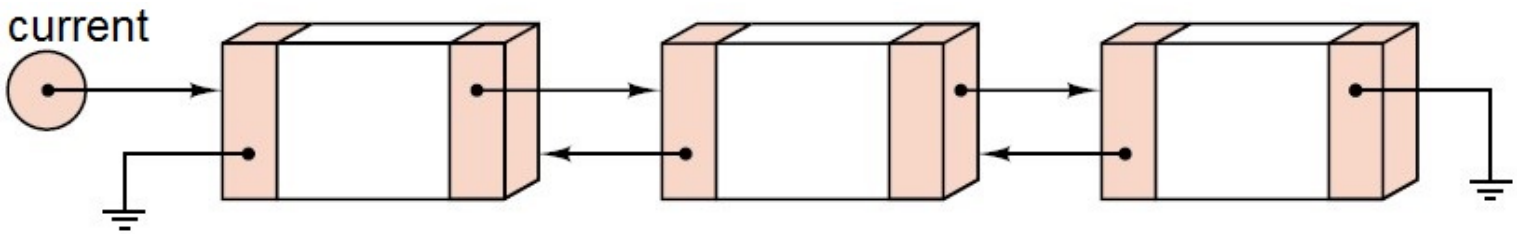


Hình: Doubly Linked List allows going forward and backward.

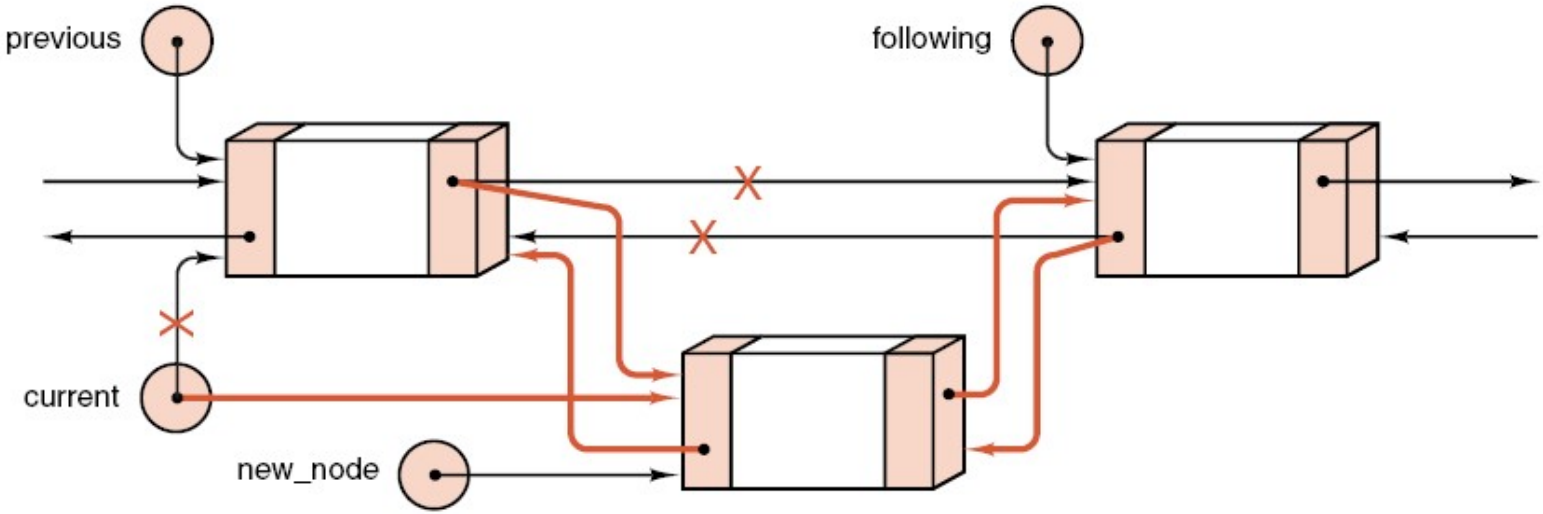
```
node
  data <dataType>
  next <pointer>
  previous <pointer>
end node
```

```
list
  current <pointer>
end list
```


Doubly Linked List

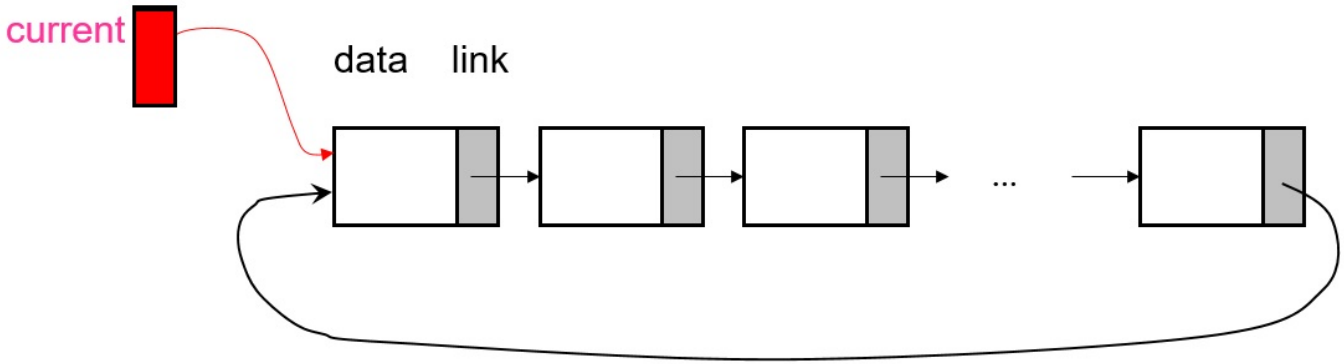


Hình: Doubly Linked List allows going forward and backward.



Hình: Insert an element in Doubly Linked List.

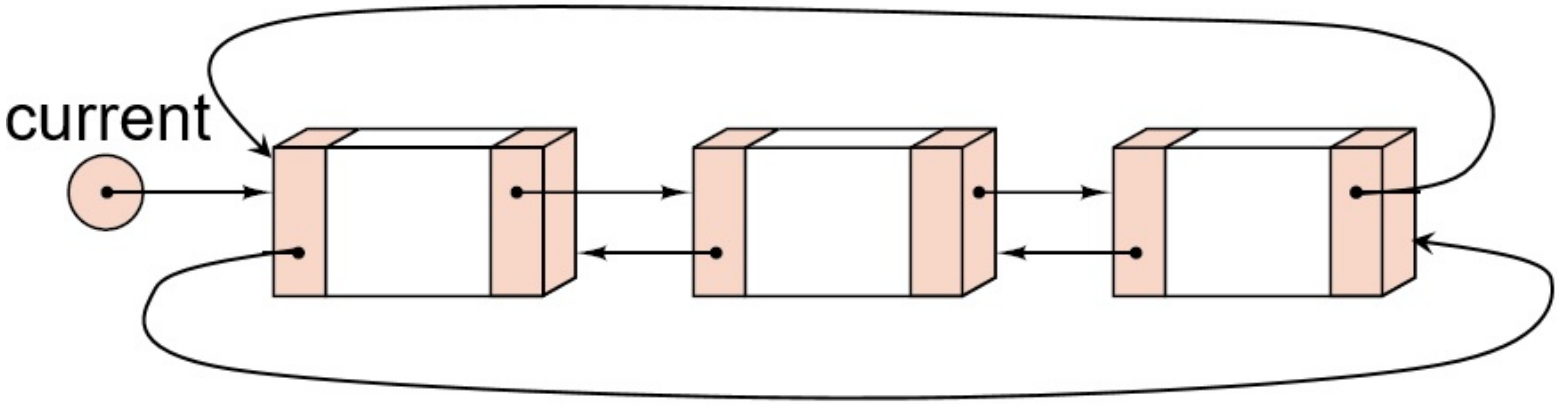
Circularly Linked List




```
node
   data <dataType>
   link <pointer>
end node
```

```
list
   current <pointer>
end list
```

Double circularly Linked List




```
node
   data <dataType>
   next <pointer>
   previous <pointer>
end node
```

```
list
   current <pointer>
end list
```

Comparison of implementations of list

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Arrays: Pros and Cons

- **Pros**:
  - Access to an array element is fast since we can compute its location quickly.
- **Cons**:
  - If we want to insert or delete an element, we have to shift subsequent elements which slows our computation down.
  - We need a large enough block of memory to hold our array.

Linked Lists: Pros and Cons

- **Pros**:
  - Inserting and deleting data does not require us to move/shift subsequent data elements.
- **Cons**:
  - If we want to access a specific element, we need to traverse the list from the head of the list to find it which can take longer than an array access.

Comparison of implementations of list

- **Contiguous storage is generally preferable when**:
  - the entries are individually very small;
  - the size of the list is known when the program is written;
  - few insertions or deletions need to be made except at the end of the list; and
  - random access is important.

- **Linked storage proves superior when**:
  - the entries are large;
  - the size of the list is not known in advance; and
  - flexibility is needed in inserting, deleting, and rearranging the entries.